The Work of a Meal

Source guide · 3 November 2026

Thirty-seven works support the essay. This guide explains what each contributes and where the evidence stops. The numbers match the online references; number 11 is unused after an unverified illustration was removed.

Read the essay: hktcr.github.io/explorations/digestion/

How to read the evidence

Primary research reports original observations. Institutional references and the textbook supply broader physiological context. Abstract-only access limits the detail that can be checked; an abstract does not establish unexamined methods. Bibliographic identity checks verify the work and citation, not the truth of every scientific claim. Unknown interests are recorded as unknown.

The essay distinguishes humans, animal preparations and laboratory assays. Its historical accounts are attributed to their authors. No fixed nutrient-uptake percentage, dietary prescription or inevitable evolutionary sequence follows from the evidence presented.

01 · Food on the International Space Station

Uri, John (2020). Food on the International Space Station. NASA, 14 August; page updated 13 January 2026.

Contribution and limits. A retrospective NASA history supports the applesauce and early swallowing observations. It is not a complete assessment of nutrient absorption in space.

Material read. selected source text.

Interests and support. Institutional editorial context; individual conflict-of-interest declarations were not supplied.

Source link

02 · Your Digestive System & How It Works

NIDDK. Your Digestive System & How It Works. Reviewed December 2017.

Contribution and limits. An institutional overview supports movement, organ functions and waste routes. Detailed cellular mechanisms are supported by the textbook and primary experiments.

Material read. selected source text.

Interests and support. Institutional editorial context; individual conflict-of-interest declarations were not supplied.

Source link

03 · Anatomy and Physiology 2e

Betts, J. Gordon, et al. (2022). Anatomy and Physiology 2e. OpenStax, Rice University. Selected sections listed below.

Contribution and limits. Textbook framework for chemistry, anatomy and transport. Relevant sections: 2.4, 2.5, 23.1, 23.2, 23.4, 23.5, 23.7, 24.1, 24.5 and 24.6. These are chapters of one work.

Material read. selected source text.

Interests and support. Institutional editorial context; individual conflict-of-interest declarations were not supplied.

Source link

Textbook sections: 2.4, 2.5, 23.1, 23.2, 23.4, 23.5, 23.7, 24.1, 24.5, 24.6

04 · Physiology of Digestion

Pavlov, Ivan. 1904. Physiology of Digestion. Nobel Lecture.

Contribution and limits. The historical lecture is evidence for Pavlov's described preparations and interpretations. His assurances about animal welfare are attributed to him, not independently certified.

Material read. selected source text.

Interests and support. A historical author's account, interpreted with its stated perspective; modern disclosure conventions do not apply.

Source link

05 · Cephalic phase of gastric secretion in healthy subjects and duodenal ulcer patients: role of vagal innervation

S J Konturek, N Kwiecien, W Obtulowicz, E Mikos, E Sito, J Oleksy, T Popiela (1979). Cephalic phase of gastric secretion in healthy subjects and duodenal ulcer patients: role of vagal innervation. Gut 20(10): 875-881. DOI 10.1136/gut.20.10.875

Contribution and limits. Main abstract results only: modified sham feeding and gastric acid/pepsin secretion in the studied healthy and duodenal-ulcer groups.

Material read. abstract.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

06 · Studies on the role of cephalic-vagal stimulation in the acid secretory response to eating in normal human subjects

C T Richardson, J H Walsh, K A Cooper, M Feldman, J S Fordtran (1977). Studies on the role of cephalic-vagal stimulation in the acid secretory response to eating in normal human subjects. Journal of Clinical Investigation 60(2): 435-441. DOI 10.1172/JCI108793

Contribution and limits. Main abstract result about gastric distension and the acid response; no universal fraction of a meal response is inferred.

Material read. abstract.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

07 · Experiments and observations on the gastric juice, and the physiology of digestion

Beaumont, William. 1833. Experiments and observations on the gastric juice, and the physiology of digestion. Plattsburgh: F. P. Allen.

Contribution and limits. Beaumont's own account supplies the injury, procedures, observations and descriptions of St Martin's choices. It does not establish unrecorded consent or private feelings.

Material read. selected source text.

Interests and support. A historical author's account, interpreted with its stated perspective; modern disclosure conventions do not apply.

Source link

08 · Working Ethics: William Beaumont, Alexis St. Martin, and Medical Research in Antebellum America

Alexa Green (2010). Working Ethics: William Beaumont, Alexis St. Martin, and Medical Research in Antebellum America. Bulletin of the History of Medicine 84(2): 193-216. DOI 10.1353/bhm.0.0341

Contribution and limits. The abstract supports the broad labour and service framing. Contract details were not examined.

Material read. abstract.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

09 · Insectivorous Plants

Darwin, Charles. 1875. Insectivorous Plants. London: John Murray. Chapter VI.

Contribution and limits. Chapter VI supplies the sundew experiments, comparisons, neutralisation trials and Darwin's interpretations. Functional analogy to pepsin is not molecular identity.

Material read. selected source text.

Interests and support. A historical author's account, interpreted with its stated perspective; modern disclosure conventions do not apply.

Source link

10 · Digestive enzymes in insectivorous plants. 3. Acid proteases in the genus Nepenthes and Drosera peltata

Shizuko AMAGASE (1972). Digestive enzymes in insectivorous plants. 3. Acid proteases in the genus Nepenthes and Drosera peltata. The Journal of Biochemistry 72(1): 73-81. DOI 10.1093/oxfordjournals.jbchem.a129899

Contribution and limits. The preparation from Drosera peltata leaves and stems differs from Darwin's leaf secretion. Acid-protease activity and the Nepenthes preparation are the uses made here.

Material read. selected source text.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

12 · The mechanisms of sodium absorption in the human small intestine

John S. Fordtran, Floyd C. Rector, Norman W. Carter (1968). The mechanisms of sodium absorption in the human small intestine. Journal of Clinical Investigation 47(4): 884-900. DOI 10.1172/JCI105781

Contribution and limits. Human small-intestinal perfusion experiment. The water-absorption result applies to the tested solutions and segments, not a fixed daily absorption share.

Material read. selected source text.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

13 · Stimulation of active and passive sodium absorption by sugars in the human jejunum

J S Fordtran (1975). Stimulation of active and passive sodium absorption by sugars in the human jejunum. Journal of Clinical Investigation 55(4): 728-737. DOI 10.1172/JCI107983

Contribution and limits. Human jejunal perfusion with specified glucose/fructose and chloride-containing solutions. The anatomy definition comes from the textbook.

Material read. selected source text.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

14 · The effect of claudin-15 deletion on cationic selectivity and transport in paracellular pathways of the cecum and large intestine

Wendy Hempstock, Nozomi Nagata, Noriko Ishizuka, Hisayoshi Hayashi (2023). The effect of claudin-15 deletion on cationic selectivity and transport in paracellular pathways of the cecum and large intestine. Scientific Reports 13(1): 6799. DOI 10.1038/s41598-023-33431-5

Contribution and limits. Main abstract result from mouse preparations: claudin-15 deletion and sodium permeability in the caecum and middle colon. No human whole-meal measurement is claimed.

Material read. abstract.

Interests and support. The indexed article record states that the authors declare no competing interests.

Source link

15 · A treatise of the scurvy, in three parts

Lind, James (1753). A treatise of the scurvy, in three parts. Edinburgh: Sands, Murray and Cochran, for A. Kincaid and A. Donaldson. Digitised copy: b30507054.

Contribution and limits. The cited transcription is the Kincaid/Donaldson 1753 copy, Internet Archive b30507054. Lind's reported allocation and successive recovery endpoints are preserved; the account is not recast as a modern randomised trial.

Material read. selected source text.

Interests and support. A historical author's account, interpreted with its stated perspective; modern disclosure conventions do not apply.

Source link

16 · Regulation of collagen synthesis by ascorbic acid

S Murad, D Grove, K A Lindberg, G Reynolds, A Sivarajah, S R Pinnell (1981). Regulation of collagen synthesis by ascorbic acid. Proceedings of the National Academy of Sciences 78(5): 2879-2882. DOI 10.1073/pnas.78.5.2879

Contribution and limits. Cultured fibroblast study, used at abstract level for the reported effect of ascorbate on collagen synthesis. It is not Lind's clinical experiment or a dosing recommendation.

Material read. abstract.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

17 · Carbohydrate fermentation in the human colon and its relation to acetate concentrations in venous blood

E W Pomare, W J Branch, J H Cummings (1985). Carbohydrate fermentation in the human colon and its relation to acetate concentrations in venous blood. Journal of Clinical Investigation 75(5): 1448-1454. DOI 10.1172/JCI111847

Contribution and limits. Human blood-acetate responses to lactulose and pectin, as reported in the abstract. Different time courses do not establish different fermentation products or a universal fibre-energy value.

Material read. abstract.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

18 · The Urinary Tract & How It Works

NIDDK. The Urinary Tract & How It Works. Reviewed June 2020.

Contribution and limits. Institutional reference for kidney waste removal and water regulation; urine is distinguished from unabsorbed intestinal residue.

Material read. selected source text.

Interests and support. Institutional editorial context; individual conflict-of-interest declarations were not supplied.

Source link

19 · The Expensive-Tissue Hypothesis: The Brain and the Digestive System in Human and Primate Evolution

Leslie C. Aiello, Peter Wheeler (1995). The Expensive-Tissue Hypothesis: The Brain and the Digestive System in Human and Primate Evolution. Current Anthropology 36(2): 199-221. DOI 10.1086/204350

Contribution and limits. A hypothesis and its authors' reply, used as an attributed explanation. It is not direct evidence of an ancestral evolutionary sequence.

Material read. selected source text.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

20 · Energetics and the evolution of human brain size

Ana Navarrete, Carel P. van Schaik, Karin Isler (2011). Energetics and the evolution of human brain size. Nature 480(7375): 91-93. DOI 10.1038/nature10629

Contribution and limits. Main abstract result from a mammalian comparison adjusted for fat-free body mass. Lack of the predicted association does not rule out every energetic trade-off.

Material read. abstract.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

21 · Artificial Selection on Relative Brain Size in the Guppy Reveals Costs and Benefits of Evolving a Larger Brain

Alexander Kotrschal, Björn Rogell, Andreas Bundsen, Beatrice Svensson, Susanne Zajitschek, Ioana Brännström, Simone Immler, Alexei A. Maklakov, Niclas Kolm (2013). Artificial Selection on Relative Brain Size in the Guppy Reveals Costs and Benefits of Evolving a Larger Brain. Current Biology 23(2): 168-171. DOI 10.1016/j.cub.2012.11.058

Contribution and limits. Guppy selection experiment: gut size, first-clutch offspring and the stated learning task. Female and male results are distinguished; this is not a direct human-ancestry test.

Material read. selected source text.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

22 · Metabolic acceleration and the evolution of human brain size and life history

Herman Pontzer, Mary H. Brown, David A. Raichlen, Holly Dunsworth, Brian Hare, Kara Walker, Amy Luke, Lara R. Dugas, Ramon Durazo-Arvizu, Dale Schoeller, Jacob Plange-Rhule, Pascal Bovet, Terrence E. Forrester, Estelle V. Lambert, Melissa Emery Thompson, Robert W. Shumaker, Stephen R. Ross (2016). Metabolic acceleration and the evolution of human brain size and life history. Nature 533(7603): 390-392. DOI 10.1038/nature17654

Contribution and limits. Living humans and apes, doubly labelled water and adjusted total expenditure. Captivity/sanctuary sampling and reused observations limit ancestral inference.

Material read. selected source text.

Interests and support. The article declares no competing financial interests; public/academic/foundation support is reported.

Source link

23 · Energetic consequences of thermal and nonthermal food processing

Rachel N. Carmody, Gil S. Weintraub, Richard W. Wrangham (2011). Energetic consequences of thermal and nonthermal food processing. Proceedings of the National Academy of Sciences 108(48): 19199-19203. DOI 10.1073/pnas.1112128108

Contribution and limits. Short feeding experiments in adult male mice. Cooked meat was associated with less body-mass loss, not a gain under these conditions.

Material read. selected source text.

Interests and support. The authors declare no conflict of interest in the paper.

Source link

24 · Digestibility of Cooked and Raw Egg Protein in Humans as Assessed by Stable Isotope Techniques

Pieter Evenepoel, Benny Geypens, Anja Luypaerts, Martin Hiele, Yvo Ghoos, Paul Rutgeerts (1998). Digestibility of Cooked and Raw Egg Protein in Humans as Assessed by Stable Isotope Techniques. The Journal of Nutrition 128(10): 1716-1722. DOI 10.1093/jn/128.10.1716

Contribution and limits. Abstract-level comparison of cooked and raw labelled egg protein in five people with ileostomies. Ileal digestibility is not whole-diet energy yield.

Material read. abstract.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

25 · Testing the expensive-tissue hypothesis' prediction of inter-tissue competition using causal modelling with latent variables

Meghan Shirley Bezerra, Samuli Helle, Kiran K. Seunarine, Owen J. Arthurs, Simon Eaton, Jane E. Williams, Chris A. Clark, Jonathan C. K. Wells (2024). Testing the expensive-tissue hypothesis' prediction of inter-tissue competition using causal modelling with latent variables. Evolutionary Human Sciences 6: e33. DOI 10.1017/ehs.2024.26

Contribution and limits. Contemporary observational modelling in seventy South Asian women. Gut size was not directly measured, and the reported estimates were inconclusive.

Material read. selected source text.

Interests and support. All eight authors declare no conflicts; public/academic/foundation support is reported.

Source link

26 · Vitamin C

MedlinePlus Medical Encyclopedia. Vitamin C. U.S. National Library of Medicine platform.

Contribution and limits. Institutional reference for vitamin C, scurvy and collagen-related functions. No intake prescription is made.

Material read. selected source text.

Interests and support. The page includes a commercial medical-content contributor; no specific product conflict was established in the material read.

Source link

27 · Direct assessment of liver glycogen storage by 13C nuclear magnetic resonance spectroscopy and regulation of glucose homeostasis after a mixed meal in normal subjects

R Taylor, I Magnusson, D L Rothman, G W Cline, A Caumo, C Cobelli, G I Shulman (1996). Direct assessment of liver glycogen storage by 13C nuclear magnetic resonance spectroscopy and regulation of glucose homeostasis after a mixed meal in normal subjects. Journal of Clinical Investigation 97(1): 126-132. DOI 10.1172/JCI118379

Contribution and limits. Human liver-glycogen measurements after a specified liquid meal, with separate related protocols. The rise and later fall are not a universal storage fraction.

Material read. selected source text.

Interests and support. The paper acknowledges an assay-standard gift from McNeil Pharmaceuticals. The workflow quality category is reduced for this disclosed material support.

Source link

28 · Intestinal transport of dipeptides in man: relative importance of hydrolysis and intact absorption

Siamak A. Adibi (1971). Intestinal transport of dipeptides in man: relative importance of hydrolysis and intact absorption. Journal of Clinical Investigation 50(11): 2266-2275. DOI 10.1172/JCI106724

Contribution and limits. Selected dipeptides in human intestinal perfusion. Intact uptake is the paper's interpretation; the study is not a complete egg meal. The original byline names Adibi as author and Morse as technical assistant.

Material read. selected source text.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

29 · Types of Ostomy Surgery of the Bowel

NIDDK. Types of Ostomy Surgery of the Bowel.

Contribution and limits. Institutional anatomical definition of an ileostomy, used to make the egg-protein study's endpoint intelligible.

Material read. selected source text.

Interests and support. Institutional editorial context; individual conflict-of-interest declarations were not supplied.

Source link

30 · Intestinal absorption of sucrose in man: interrelation of hydrolysis and monosaccharide product absorption

G M Gray, F J Ingelfinger (1966). Intestinal absorption of sucrose in man: interrelation of hydrolysis and monosaccharide product absorption. Journal of Clinical Investigation 45(3): 388-398. DOI 10.1172/JCI105354

Contribution and limits. Human intestinal sucrose perfusion compared with free glucose/fructose. Remaining luminal products distinguish hydrolysis from uptake; the substrate was sucrose, not bread starch.

Material read. selected source text.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

31 · THE MICROBIAL CONTRIBUTION TO HUMAN FAECAL MASS

A. M. Stephen, J. H. Cummings (1980). THE MICROBIAL CONTRIBUTION TO HUMAN FAECAL MASS. Journal of Medical Microbiology 13(1): 45-56. DOI 10.1099/00222615-13-1-45

Contribution and limits. The abstract reports bacteria separated from human faeces. Only bacterial presence is used; the measured proportion in the small dietary sample is not generalised.

Material read. abstract.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

32 · Quantitation of Hepatic Glycogenolysis And Gluconeogenesis in Fasting Humans With 13C NMR

Douglas L. Rothman, Inger Magnusson, Lee D. Katz, Robert G. Shulman, Gerald I. Shulman (1991). Quantitation of Hepatic Glycogenolysis And Gluconeogenesis in Fasting Humans With 13C NMR. Science 254(5031): 573-576. DOI 10.1126/science.1948033

Contribution and limits. Human fasting study using serial liver-glycogen measurements and glucose-production estimates. Supports a contribution from glycogen breakdown, not a constant proportion or tracking of this essay's imagined meal.

Material read. abstract.

Interests and support. No article-specific declaration was available in the material read; interests remain unknown.

Source link

33 · Glucose/galactose malabsorption caused by a defect in the Na+/glucose cotransporter

E. Turk, B. Zabel, S. Mundlos, J. Dyer, E. M. Wright (1991). Glucose/galactose malabsorption caused by a defect in the Na+/glucose cotransporter. Nature 350(6316): 354-356. DOI 10.1038/350354a0

Contribution and limits. A human family with glucose/galactose malabsorption, with functional testing of the altered transporter in frog oocytes. Abstract-level evidence; no separate galactose transport assay is claimed.

Material read. abstract.

Interests and support. UCLA physiology affiliation and government research support visible. No article-specific COI declaration recovered; unknown, not absent.

Source link

34 · Human intestinal H+/peptide cotransporter. Cloning, functional expression, and chromosomal localization

Rong Liang, You-Jun Fei, Puttur D. Prasad, Sammanda Ramamoorthy, Hong Han, Teresa L. Yang-Feng, Matthias A. Hediger, Vadivel Ganapathy, Frederick H. Leibach (1995). Human intestinal H+/peptide cotransporter. Cloning, functional expression, and chromosomal localization. Journal of Biological Chemistry 270(12): 6456-6463. DOI 10.1074/jbc.270.12.6456

Contribution and limits. A cloned human intestinal transporter expressed in cultured cells and frog oocytes. Supports di-/tripeptide transport, not free-amino-acid uptake or subsequent intracellular peptide cleavage.

Material read. abstract.

Interests and support. Medical College of Georgia academic affiliation and NIH/NIDDK grant visible. Article-specific COI declaration not recovered; unknown.

Source link

35 · Structural insight into substrate specificity of human intestinal maltase-glucoamylase

Limei Ren, Xiaohong Qin, Xiaofang Cao, Lele Wang, Fang Bai, Gang Bai, Yuequan Shen (2011). Structural insight into substrate specificity of human intestinal maltase-glucoamylase. Protein & Cell 2(10): 827-836. DOI 10.1007/s13238-011-1105-3

Contribution and limits. Structural and biochemical work with human maltase-glucoamylase. The primary result concerns active-site structure and substrate preference; textbook physiology supports the full brush-border account.

Material read. abstract.

Interests and support. Nankai University affiliations visible; drug-target motivation is not itself evidence of a financial conflict. Article-specific declaration unavailable; unknown.

Source link

36 · Absorption of oleic and palmitic acids from emulsions and micellar solutions

W. J. Simmonds, T. G. Redgrave, R. L. S. Willix (1968). Absorption of oleic and palmitic acids from emulsions and micellar solutions. Journal of Clinical Investigation 47(5): 1015-1025. DOI 10.1172/JCI105791

Contribution and limits. Controlled fatty-acid infusions in rats with thoracic-duct lymph collection. Labelled-fatty-acid transfer and esterified-fat output are distinguished. Micelles were not indispensable in this preparation.

Material read. selected source text.

Interests and support. Public/university support and a medical-research fellowship are acknowledged. Wyandotte Chemical Co. supplied Pluronic F68, a tested comparator; the workflow quality category is reduced for this material support.

Source link

37 · Pancreatic lipase assays with triglycerides as substrate: contribution of each sequential reaction to product formation

A Lykidis, V Mougios, P Arzoglou (1994). Pancreatic lipase assays with triglycerides as substrate: contribution of each sequential reaction to product formation. Clinical Chemistry 40(11): 2053-2056. DOI 10.1093/clinchem/40.11.2053

Contribution and limits. Human pancreatic-lipase assays in vitro. The abstract supports sequential glyceride hydrolysis; assay fractions and timing are not transferred to an ordinary meal.

Material read. abstract.

Interests and support. University chemistry affiliation inspected; full-paper funding/COI not accessible, hence unknown, no identified relevant COI.

Source link

38 · Electron microscopic studies of the assembly, intracellular transport, and secretion of chylomicrons by rat intestine

S M Sabesin, S Frase (1977). Electron microscopic studies of the assembly, intracellular transport, and secretion of chylomicrons by rat intestine. Journal of Lipid Research 18(4): 496-511. DOI 10.1016/S0022-2275(20)41667-0

Contribution and limits. An indexed original abstract excerpt describes ultrastructural work in corn-oil-fed rats. It supports chylomicron assembly/secretion in that preparation, not the entire onward lymph-to-blood pathway.

Material read. abstract excerpt.

Interests and support. Not available in actual excerpt; formal COI/funding unknown.

Source link